Solution of the Spacewire exercice

1. D_eed(f1)= D_S1(f1)+ D_A(f1)+D_B(f1)+TR(f1)

D_S1(f1)
= D_eed (A=> S2) (f2)
= D_A(f2)+TR(f2)
=0+ L(f2)/C

Since there is no packet at the router A waiting for the same output than f2, then it is delay in the router A is null.

D_A(f1)
= max(D_eed(A=>S5) (f3), D_eed(A=>S4)(f4))
=max(D_B(f3)+TR(f3), D_B(f4)+TR(f4))
= max(D_eed(B=>S5) (f6) + TR(f3), D_eed(B=>S4)(f5)+TR(f4))
=max(0+TR(f6)+TR(f3), 0+TR(f5) +TR(f4))
= max (L(f6)/C + L(f3)/C, L(f5)/C + L(f4)/C)

There are f3 and f4 coming from the same input port and want to access the same output port than f1, thus we take the maximum of their end-to-end delays until arriving their final destination. Afterwards, we have f3, which can be blocked by flow f6; whereas f4 that can be delayed by flow f5. The flows f5 and f6 are the only flows asking for their corresponding outputs in router B; thus we count only their transmission times.

D_B(f1)
=D_eed(B=>S3)(f7)
=TR(f7)
= L(f7)/C
There is only flow 7 waiting for the same output than f1 in router B; thus the final result.

D_eed(f1) = L(f2)/C + max (L(f6)/C + L(f3)/C, L(f5)/C + L(f4)/C)+ L(f7)/C+ L(f1)/C

2. D_eed(f4) = D_S2(f4) + D_A(f4)+ D_B (F4)+ TR(f4)

D_S2(f4)
= D_eed(A=>S5)(f3)
= D_A(f3)+ D_B (f3) +TR(f3)
= D_eed(A=>S3) (f1) + D_eed(B=>S5) (f6)+ TR(f3)

= D_B(f1)+TR(f1)+ TR(f6)+ TR(f3)
= D_eed(B=>S3) (f7) +TR(f1)+TR(f6)+TR(f3)
= TR(f7) + TR(f1) + TR(f6) + TR(f3)
= L(f7)/C + L(f1)/C + L(f6)/C + L(f3)/C

D_A(f4)
= D_eed(A=>S3) (f1)
= D_B(f1)+ TR(f1)
= D_eed(B=>S3)(f7)+ TR(f1)
=TR(f7) + TR(f1)
= L(f7)/C + L(f1)/C

D_B(f4)
= D_eed(B=>S4) (f5)
=TR(f5)
= L(f5)/C

Hence ,
D_eed(f4)
= L(f7)/C + L(f1)/C + L(f6)/C + L(f3)/C + L(f7)/C + L(f1)/C + L(f5)/C + L(f4)/C

3. D_eed(f7) = D_S4(f7) + D_B (f7) + TR(f7)

D_S4(f7)
= D_eed (B=>S5)(f6)
= D_B(f3)+ TR(f6)
= L(f3)/C + L(f6)/C

D_B(f7)
= D_eed(B=>S3)(f1)
= TR(f1)
= L(f1)/C

Hence,
D_eed(f7) = L(f3)/C + L(f6)/C + L(f1)/C + L(f7)/C